

ID Number: _____

Birla Vishvakarma Mahavidyalaya (Engineering College)
(An Autonomous Institution)

Third Level, B.Tech. (Computer Engineering)

End Semester Examination, Summer 2022

Course Code: 3CP06

Course Title: Theory of Computation

Date: 25.04.2022

Time: 11:00 am to 01:30 pm

Maximum Marks: 60

Instructions:

- Each section must be written in a separate answer book.
- Numbers in the square brackets to the right indicate maximum marks.
- The light-coloured parenthesized text on the right hand side of each question indicates the CO(s) addressed by that question followed by the level of that question according to Bloom's taxonomy, i.e., R: Remember, U: Understand, A: Apply, N: Analyse, E: Evaluate, and C: Create.
- Make use of neat sketches wherever possible, to keep your explanations brief, precise, and clear.
- Make necessary assumptions, stating the same clearly.

Section - I

1. (A) Define following terms: 2R [4]
 - i. Context Free Grammar (CFG)
 - ii. Zero or more step derivation relationship (i.e., $\Rightarrow^*$) for CFGs
 - iii. Deterministic Push Down Automaton (PDA)
 - iv. Acceptance of a string by a PDA by empty stack

(B) Design the asked type of Finite Automaton (FA) for the stated language: 1A [6]

 - i. Deterministic FA (DFA) for $L_1 = \{x \mid 00 \text{ is not a substring of } x\}$
 - ii. DFA for $L_2 = \{x \mid x \text{ ends with } 01\}$
 - iii. ϵ -NFA (i.e., Non-deterministic FA with ϵ transitions) for the language described by the regular expression 001^*0^*11
2. (A) Define four types of grammars (i.e., Type 0, 1, 2, and 3). 2R [4]

(B) Construct CFGs for **Any Two** of the following languages: 2A [6]

 - i. $\{x \in \{a, b\}^* \mid x \text{ has an even number of } a\text{'s followed by an even number of } b\text{'s}\}$
 - ii. $\{x \in \{a, b, c\}^* \mid x = a^i b^j c^k, \text{ where } i, j, k \geq 0 \text{ and } i = j + k\}$
 - iii. $\{x \in \{a, b\}^* \mid n_a(x) \neq n_b(x)\}$
3. (A) Answer **Any One** of the following questions: 2A [6]
 - i. Use pumping lemma for CFLs to prove that $L = \{ss \mid s \in \{a, b\}^*\}$ is not a CFL.
 - ii. Design a Push Down Automaton (PDA) to accept by final state, the language $\{x \in \{a, b\}^* \mid n_a(x) = n_b(x)\}$.
 - α) List each of the 7 components clearly. Draw transition diagram for showing δ . [4]
 - β) Starting from the initial ID (q_0, w, Z_0) , show all reachable ID's for [2]
 - (i) $w = abbaab$ and (ii) $w = aaaabb$.

(B) Convert following CFG to CNF: 3A [4]

$G = (\{A, B, S\}, \{0, 1\}, P, S)$, where $P = \{S \rightarrow ABA, A \rightarrow 0A \mid \epsilon, B \rightarrow 1B \mid \epsilon\}$

Section - II

4. (A) Consider the Turing Machine (TM)

$$M = (\{q_0, q_1, q_f\}, \{0, 1\}, \{0, 1, B\}, \delta, q_0, B, \{q_f\})$$

where the transition function δ consists of the following set of rules:

$$\delta(q_0, 0) = (q_1, 1, R); \delta(q_1, 1) = (q_0, 0, R); \delta(q_0, B) = (q_f, B, R)$$

- i. Determine $L(M)$. 5A [2]
- ii. State the output left on the tape when a string is accepted. 5A [2]
- (B) Draw a sketch of each of the followings and briefly explain how a single δ move is made in each. 5U
 - i. A four-track (on a single tape) TM [2]
 - ii. A two-tape TM [2]
- (C) Consider the description of the transition function (i.e., δ) of any TM as a quintuple (p, X, q, Y, D) , where p is the current state, X is the current tape symbol being scanned by the head, q is the next state, Y is the symbol that will replace X on the tape, and D is the direction in which the head will move (L or R), respectively. [2]
 Suppose, for a TM $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$, we have been given the set of all required quintuples that define the δ completely. If, besides the set of quintuples, we don't have any other information about M , which of the seven components of M can not be determined? (Just state the names of such components.) 5U
5. (A) Design a TM M_1 that accepts, by reaching to a final state, all *even* length palindromes over $\{a, b\}$. Draw the transition diagram of M_1 . 5A [4]
- (B) Design a TM M_2 that accepts, by reaching to a final state, the language over $\{a, b\}$ containing all strings that start with an a and end with a b . Draw the transition diagram of M_2 . 5A [4]
- (C) Starting from the TM M_2 you designed in Q.5(B), construct another TM M'_2 that accepts $L(M)$ by halting. Draw transition diagram of M'_2 . 5A [2]
6. (A) Select the *most appropriate* answer from the given options: (Write Only your selected option in your answer book.) 6U [2]
 - i. Which of the following statements are True, for $\Sigma = \{a, b\}$:
 (i) Σ^* is recursive (ii) Σ^* is recursively enumerable (re) (iii) $\forall L \subseteq \Sigma^*$, L is recursive (iv) $\forall L \subseteq \Sigma^*$, L is re
 (a) (i) and (ii) (b) (ii) and (iii) (c) (iii) and (iv) (d) (i) and (iv)
 (e) (ii) and (iv)
 - ii. Let A be the set of all *re* languages and B be the set of all recursive languages. Which of the following statements is true?
 (a) $A \subset B$ (b) $A \subseteq B$ (c) $B \subset A$ (d) $B \subseteq A$
- (B) Using systematic arguments and appropriate sketches, show that ... 6A [4]
 - i. If a language L is recursive then its complement, $\bar{L}$ is also recursive.
 - ii. If a language L and $\bar{L}$ are *re* then L is recursive.
- (C) What is 'Halting Problem'? Using systematic arguments and appropriate sketches, show that the Halting problem is undecidable. 6A [4]